

Lecture 8 Image Super Resolution

(Generative Models for Super Resolution)

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SIST Building-3 420

Outline

- Interpolation methods
- Reconstruction based methods
- Deep learning based methods
- Single-image Super Resolution



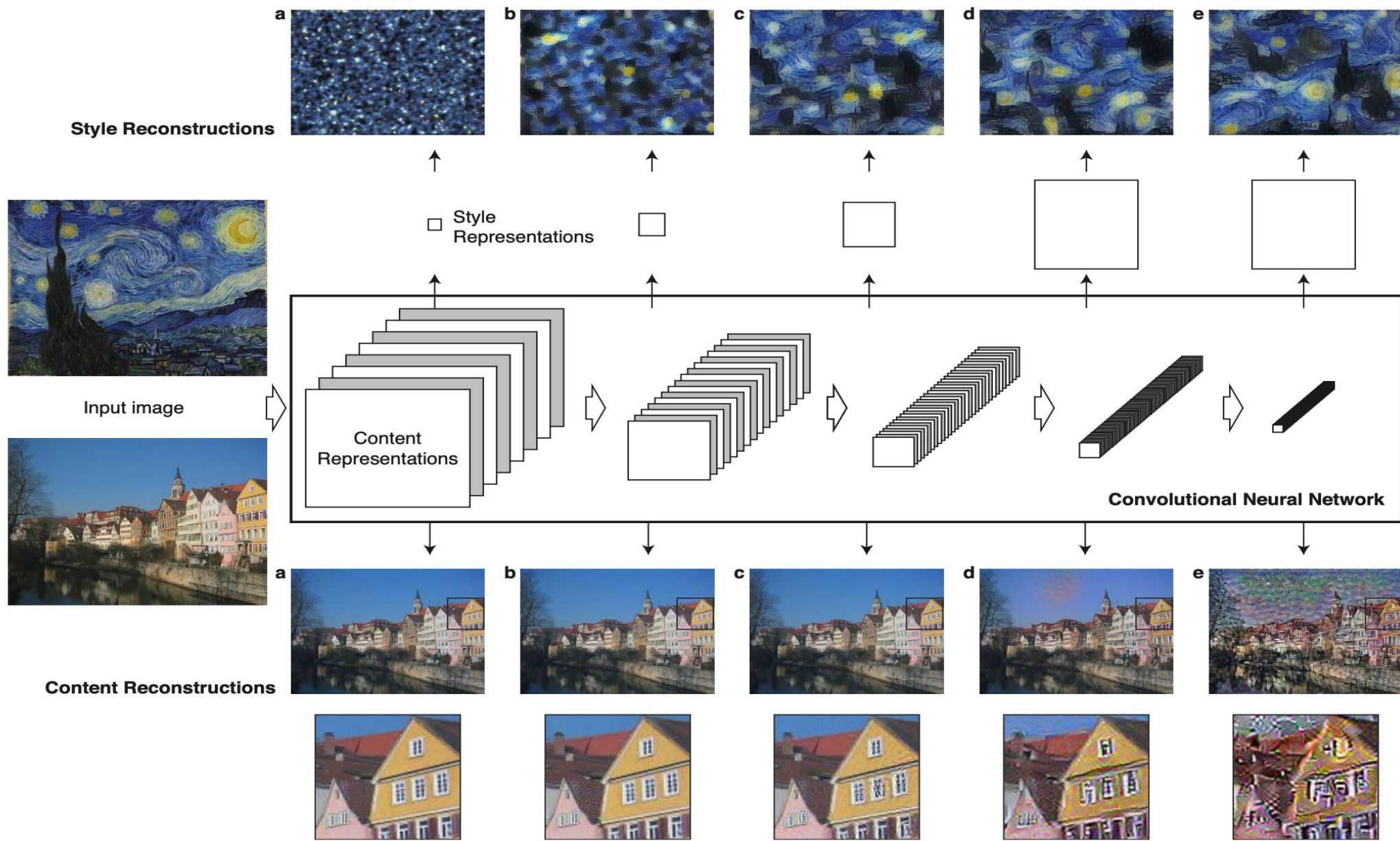
Generative Models: SRGAN & EnhanceNet & ESRGAN

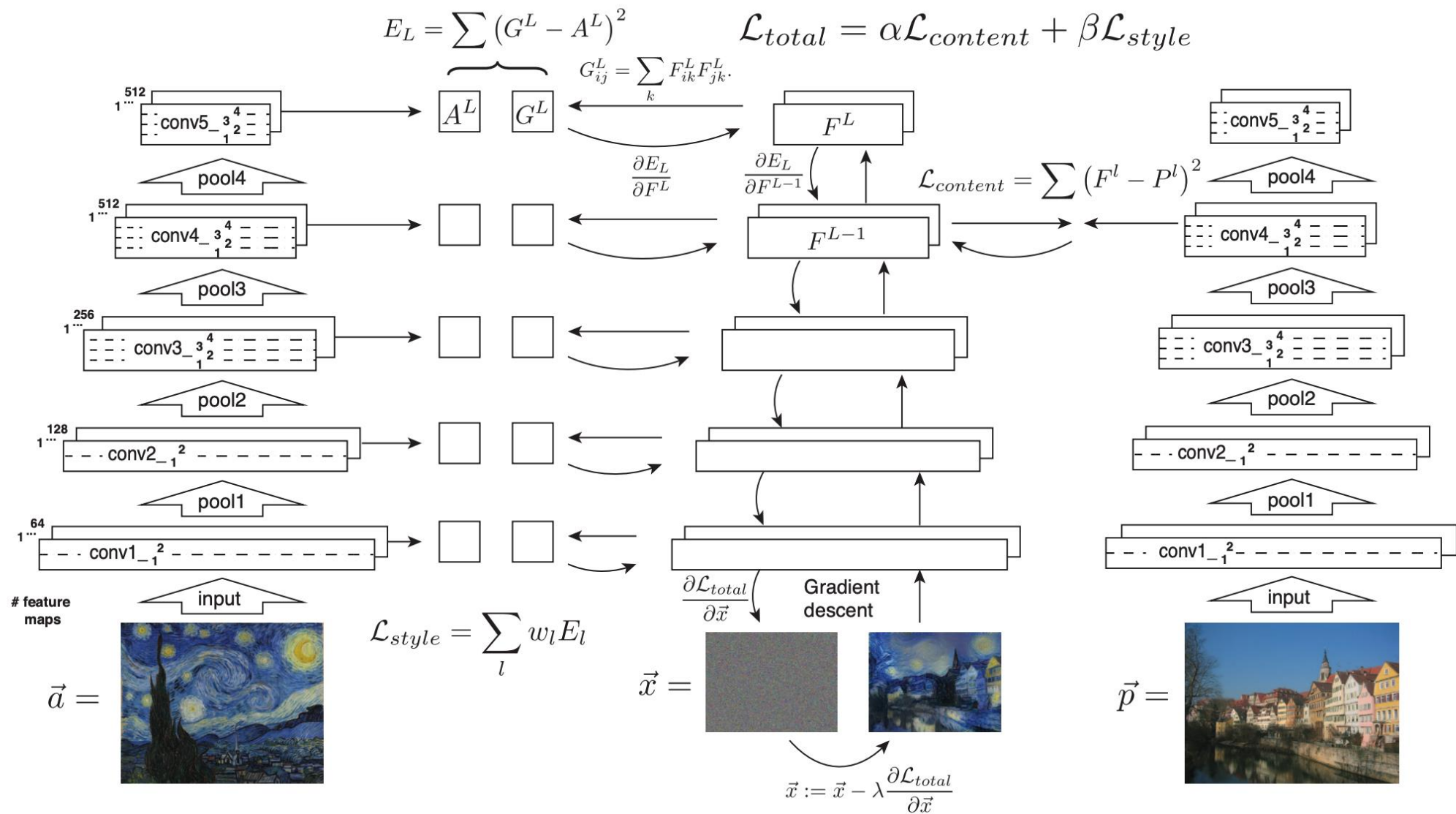
[1] Leon A. Gatys, Alexander S. Ecker, Matthias Bethge, Image Style Transfer Using Convolutional Neural Networks (CVPR 2016)

[2] Justin Johnson, Alexandre Alahi, Li Fei-Fei, Perceptual Losses for Real-Time Style Transfer and Super-Resolution (ECCV 2016)

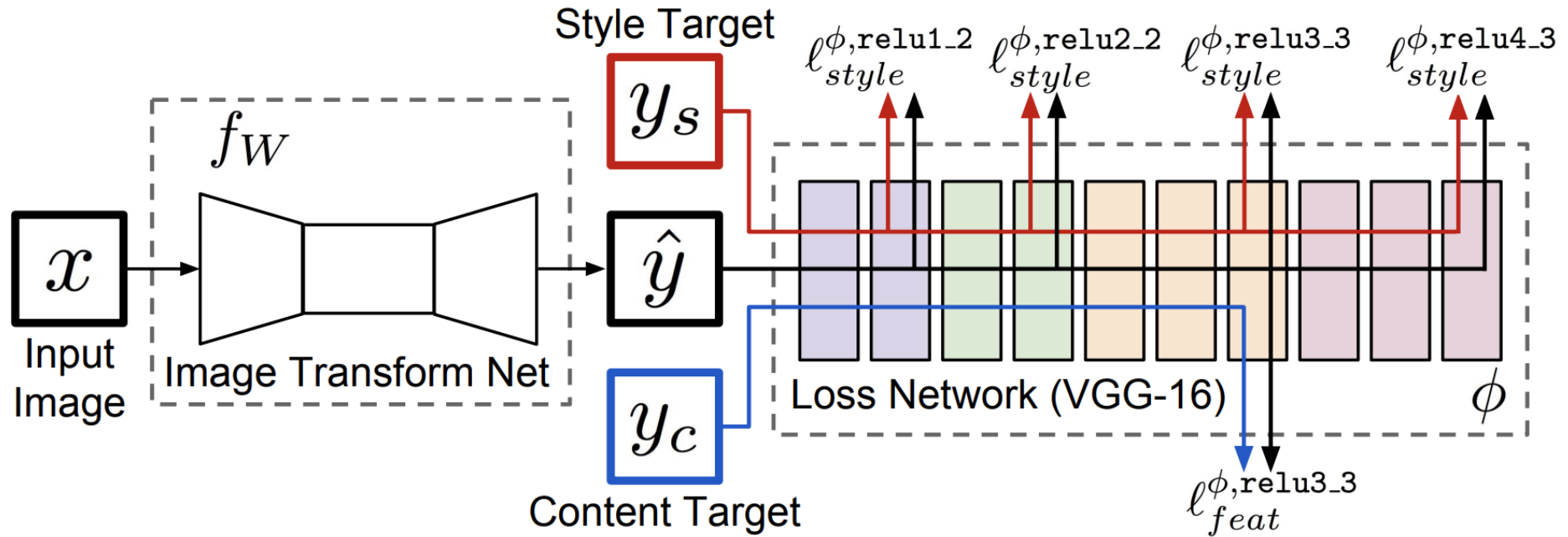
Perceptual Loss

- Feature Reconstruction Loss. Rather than encouraging the pixels of the output image $\hat{y} = f_W(x)$ to exactly match the pixels of the target image y , we instead encourage them to have similar feature representations as computed by the loss network ϕ .
- The networks discussed so far give equal importance to all spatial locations . In all our experiments ϕ is the 16-layer VGG network [46] pretrained on the ImageNet dataset.





Perceptual Loss



Feature Reconstruction Loss

$$\ell_{feat}^{\phi,j}(\hat{y}, y) = \frac{1}{C_j H_j W_j} \|\phi_j(\hat{y}) - \phi_j(y)\|_2^2$$

Style Reconstruction Loss.

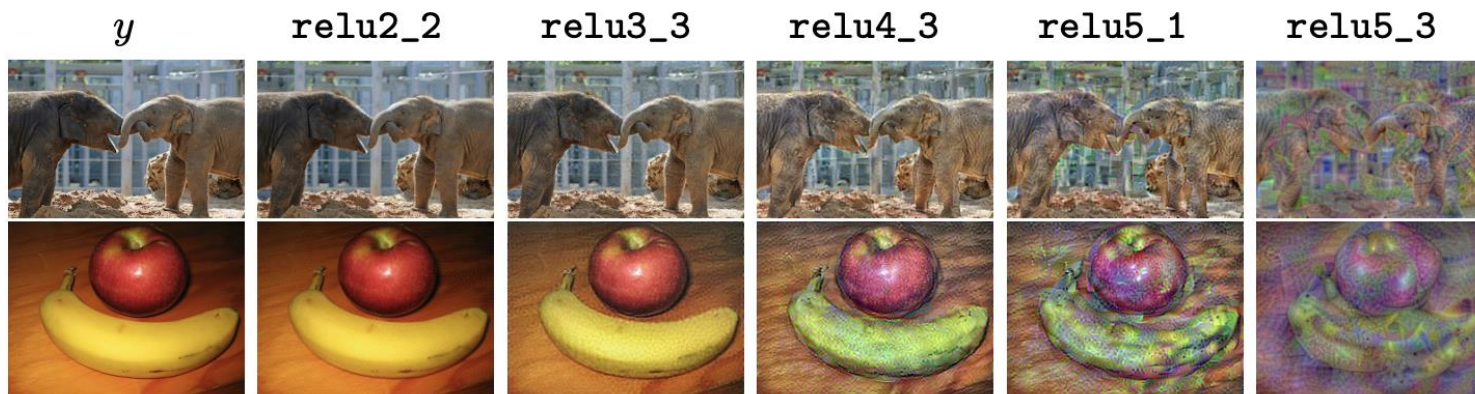
$$\ell_{style}^{\phi,j}(\hat{y}, y) = \|G_j^{\phi}(\hat{y}) - G_j^{\phi}(y)\|_F^2.$$

$$G_j^{\phi}(x)_{c,c'} = \frac{1}{C_j H_j W_j} \sum_{h=1}^{H_j} \sum_{w=1}^{W_j} \phi_j(x)_{h,w,c} \phi_j(x)_{h,w,c'}.$$

Perceptual Loss

Feature Reconstruction Loss

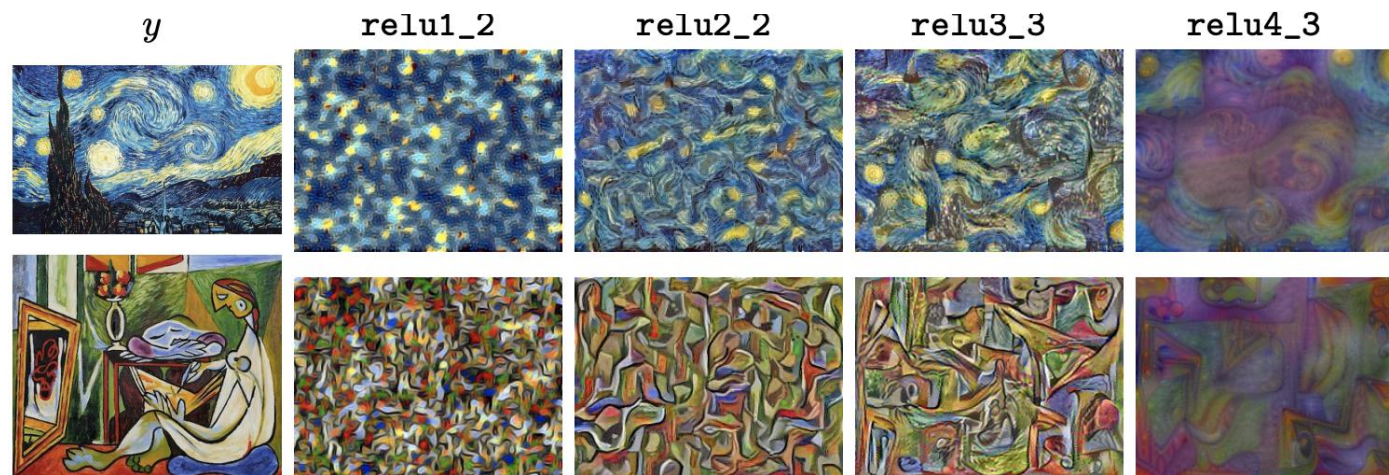
$$\ell_{feat}^{\phi,j}(\hat{y}, y) = \frac{1}{C_j H_j W_j} \|\phi_j(\hat{y}) - \phi_j(y)\|_2^2$$



Style Reconstruction Loss.

$$\ell_{style}^{\phi,j}(\hat{y}, y) = \|G_j^{\phi}(\hat{y}) - G_j^{\phi}(y)\|_F^2.$$

$$G_j^{\phi}(x) = \psi\psi^T / C_j H_j W_j.$$

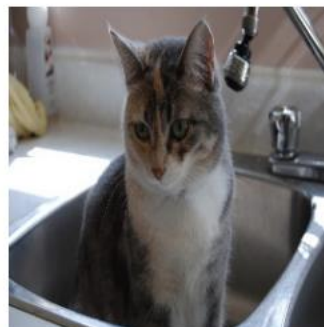


Perceptual Loss

Style
Sketch



Style
The Simpsons



Content

[10]

Ours

Content

[10]

Ours

Perceptual Loss



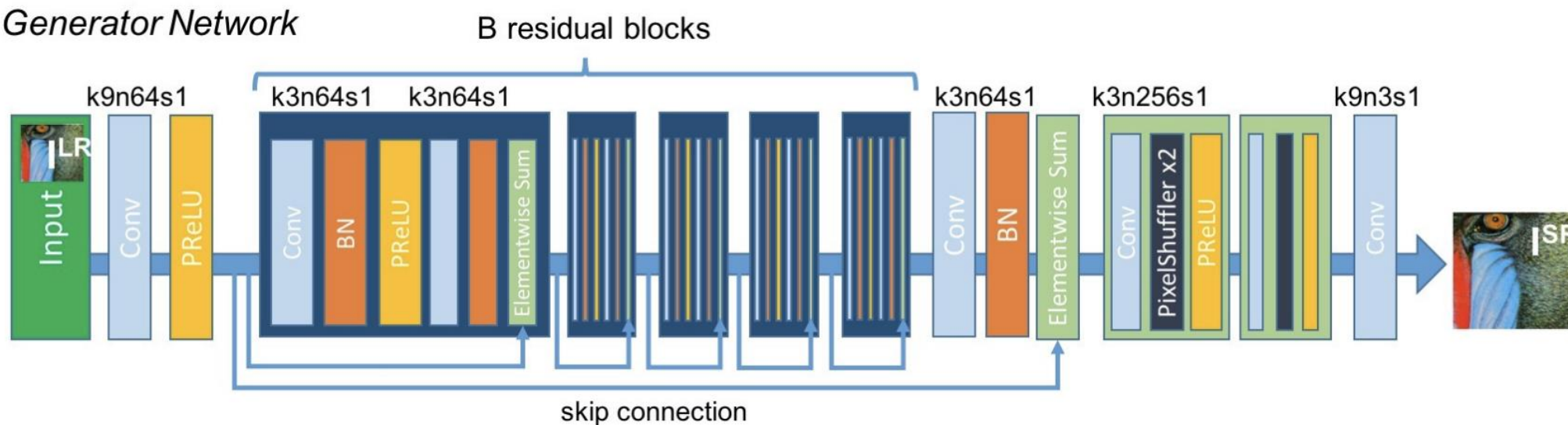
Ground Truth	Bicubic	Ours (ℓ_{pixel})	SRCNN [11]	Ours (ℓ_{feat})
This Image	21.69 / 0.5840	21.66 / 0.5881	22.53 / 0.6524	21.04 / 0.6116
Set14 mean	25.99 / 0.7301	25.75 / 0.6994	27.49 / 0.7503	24.99 / 0.6731
BSD100 mean	25.96 / 0.682	25.91 / 0.6680	26.90 / 0.7101	24.95 / 63.17

SRGAN

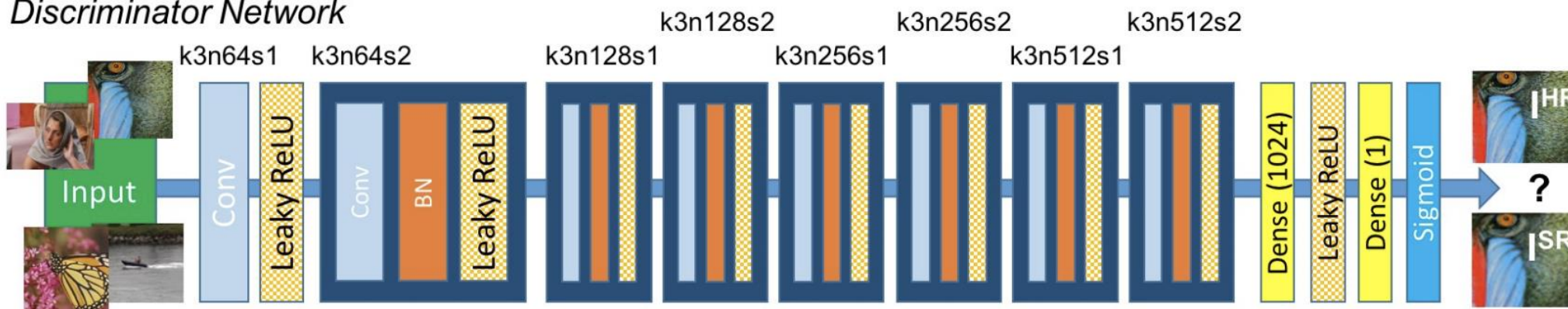
- It is the first framework capable of inferring photo-realistic natural images for $4\times$ upscaling factors.
- We propose a perceptual loss function which consists of an adversarial loss and a content loss. The adversarial loss pushes our solution to the natural image manifold using a discriminator network that is trained to differentiate between the super-resolved images and original photo-realistic images

SRGAN

Generator Network



Discriminator Network



SRGAN



Figure 2: From left to right: bicubic interpolation, deep residual network optimized for MSE, deep residual generative adversarial network optimized for a loss more sensitive to human perception, original HR image. Corresponding PSNR and SSIM are shown in brackets. [4× upscaling]

SRGAN

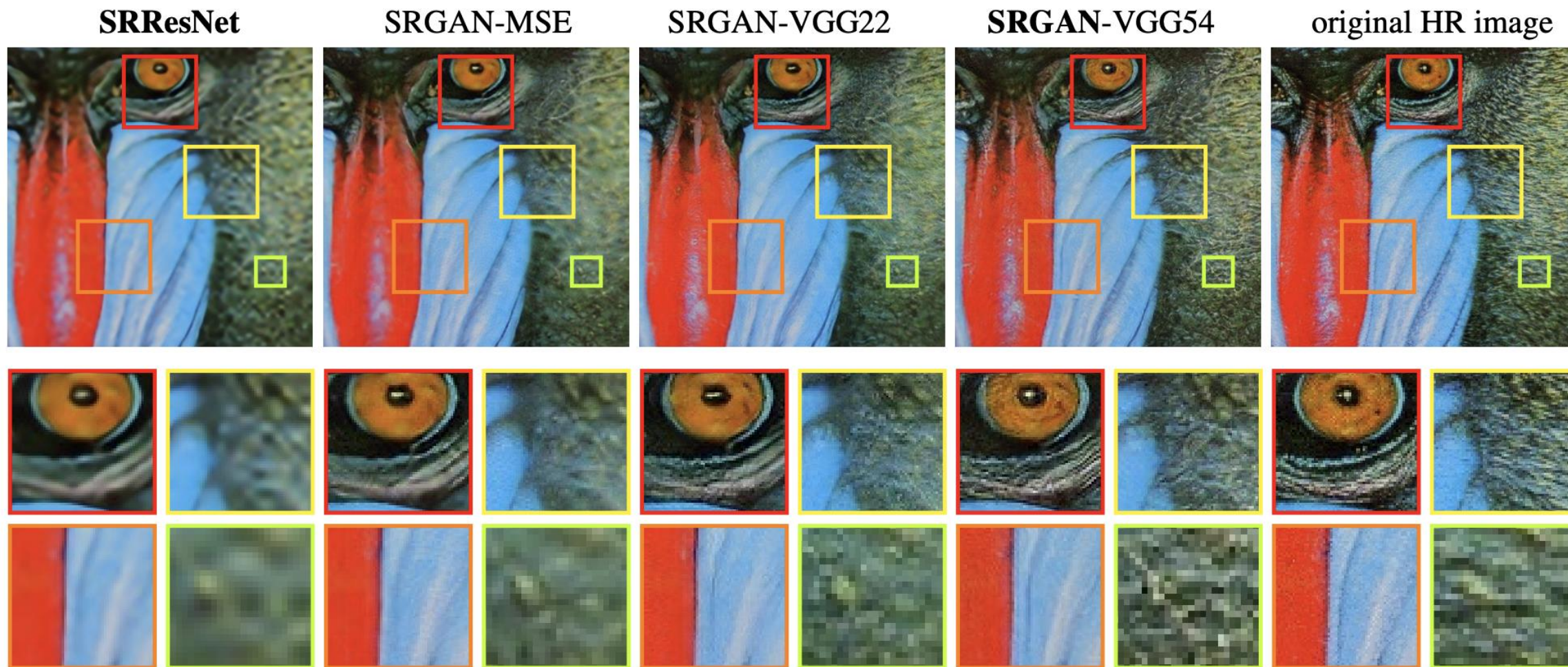
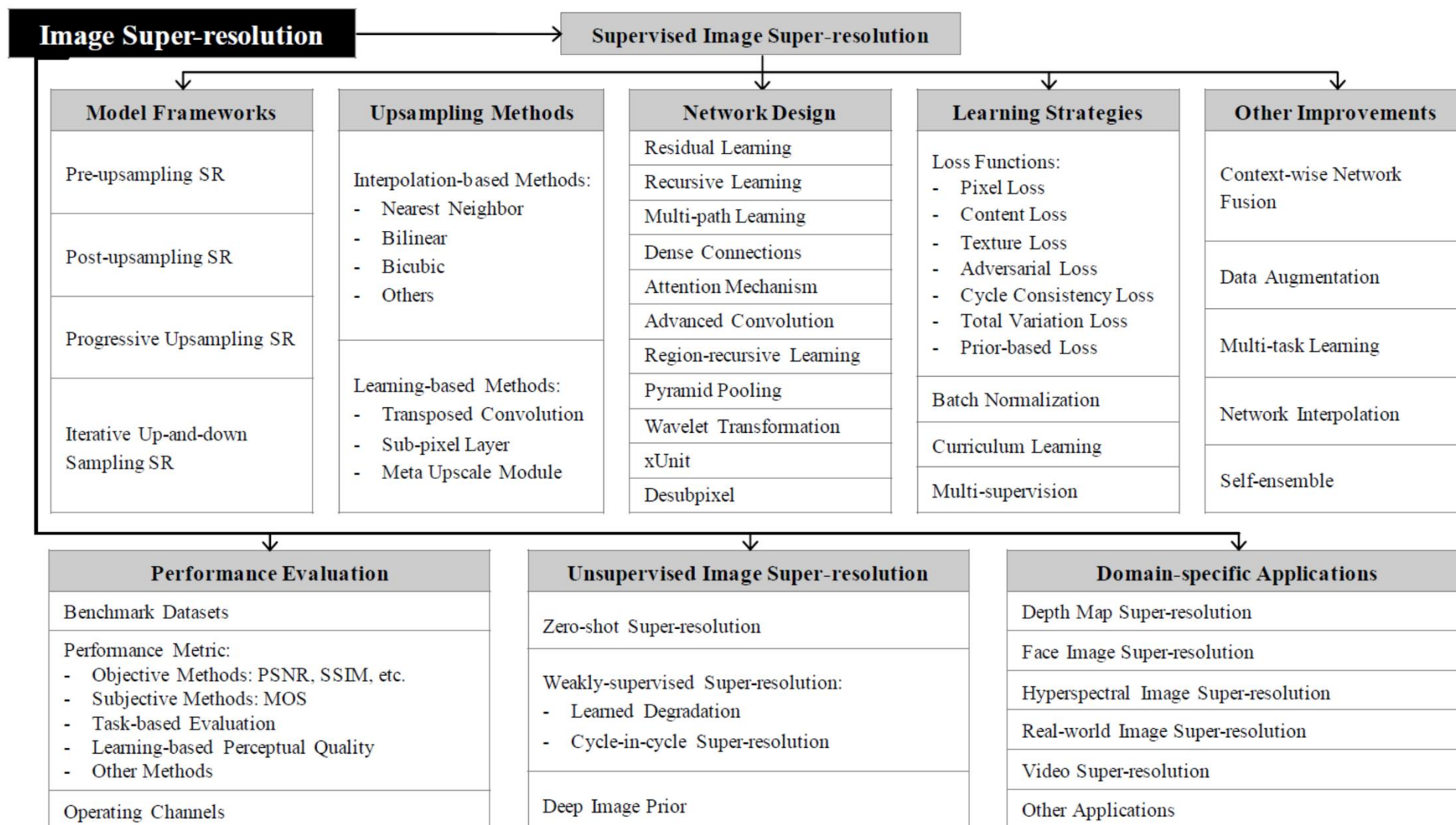


Figure 6: **SRResNet** (left: a,b), **SRGAN-MSE** (middle left: c,d), **SRGAN-VGG2.2** (middle: e,f) and **SRGAN-VGG54** (middle right: g,h) reconstruction results and corresponding reference HR image (right: i,j). [4× upscaling]

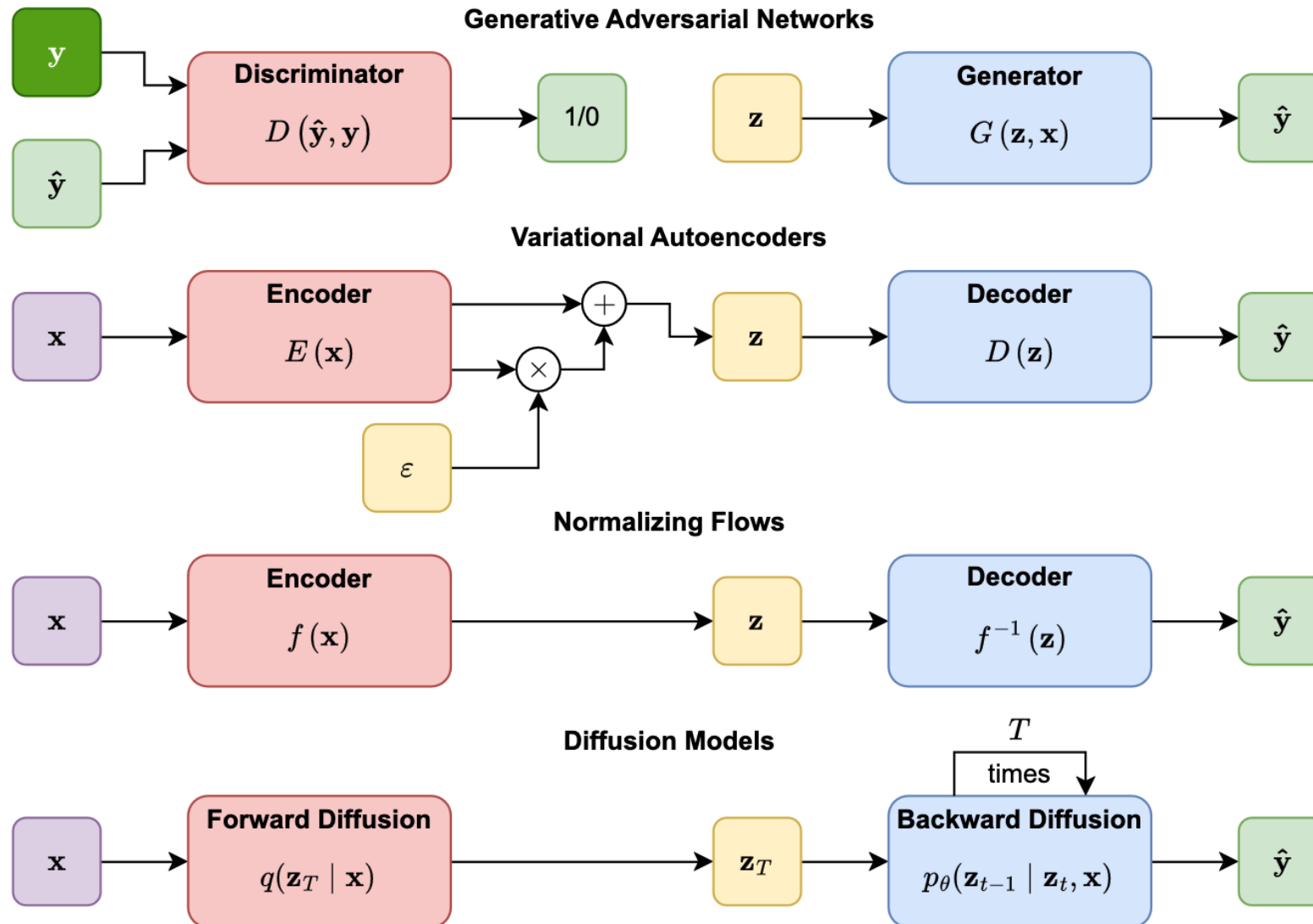
Summary



Diffusion Model based Super Resolution

- [1] SR3: Image Super-Resolution via Iterative Refinement, TPAMI 2023
- [2] Implicit Diffusion Models for Continuous Super-Resolution, CVPR 2023

Conceptual overview of generative models (GANs, VAEs, NFs, and DMs).



Variational Autoencoders

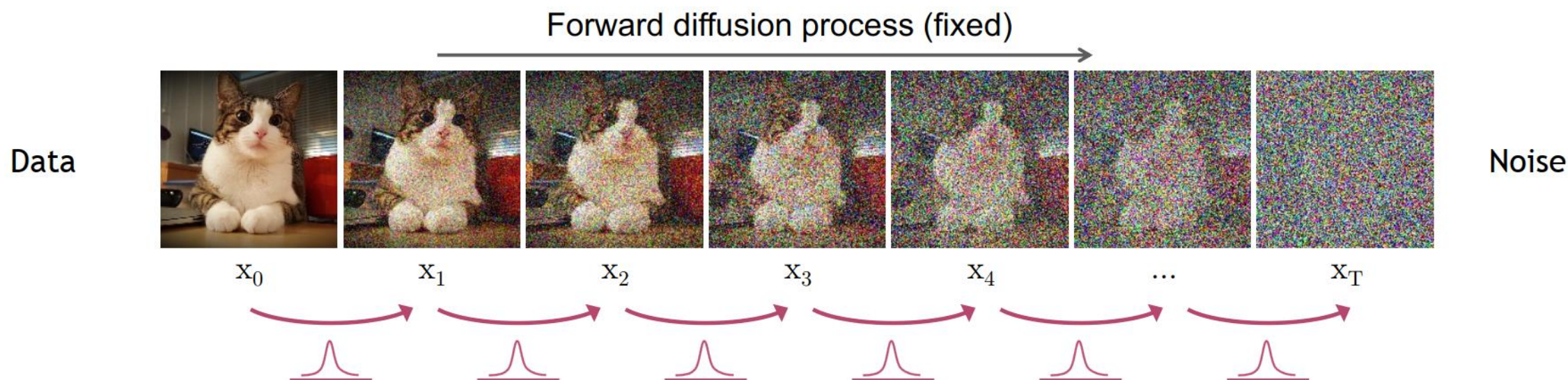
$$\begin{aligned} KL(q_\theta(z|x) \parallel p(z|x)) &= \int q_\theta(z|x) \ln \frac{q_\theta(z|x)}{p(z|x)} dz \\ &= \mathbb{E}_{z \sim q_\theta(z|x)} \left[\ln \frac{q_\theta(z|x)}{p(z|x)} \right] \\ &= \mathbb{E}_{z \sim q_\theta(z|x)} [\ln q_\theta(z|x) - \ln p(z|x)] \\ &= \mathbb{E}_{z \sim q_\theta(z|x)} \left[\ln q_\theta(z|x) - \ln \frac{p(x|z)p(z)}{p(x)} \right] \\ &= \mathbb{E}_{z \sim q_\theta(z|x)} [\ln q_\theta(z|x) - \ln p(z) - \ln p(x|z)] + \ln p(x) \\ &= KL(q_\theta(z|x) \parallel p(z)) - \mathbb{E}_{z \sim q_\theta(z|x)} [\ln p(x|z)] + \ln p(x) \end{aligned}$$

Then we have

$$\ln p(x) - KL(q_\theta(z|x) \parallel p(z|x)) = \mathbb{E}_{z \sim q_\theta(z|x)} [\ln p(x|z)] - KL(q_\theta(z|x) \parallel p(z))$$

Forward Diffusion Process

The formal definition of the forward process in T steps

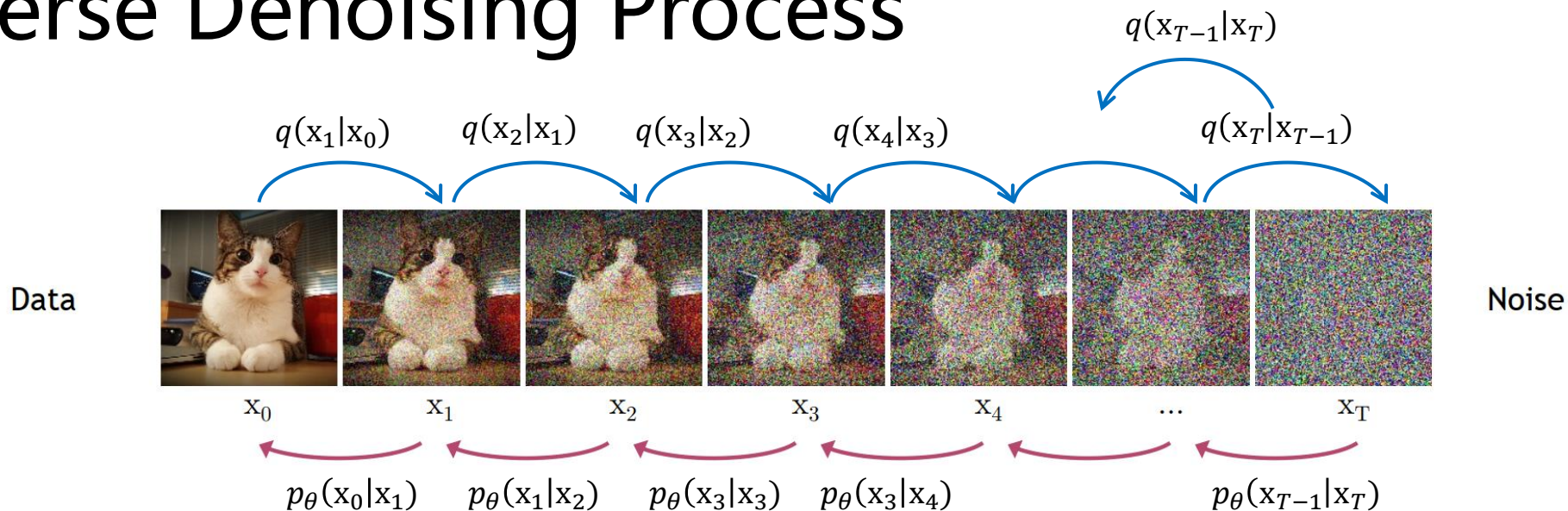


$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I) \quad \longrightarrow \quad q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1})$$

The steps sizes are controlled by a variance schedule $\{\beta_t \in (0,1)\}_{t=1}^T$



Reverse Denoising Process



Train a network to approximate $q(x_{t-1}|x_t)$

$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\mu_\theta(\mathbf{x}_t, t)}_{\text{Trainable network}}, \sigma_t^2 \mathbf{I}) \quad \rightarrow \quad p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Trainable network
(U-net, Denoising Autoencoder)



Generative Learning by Denoising

We know prior $q(x_t|x_{t-1})$, Can we get posterior $q(x_{t-1}|x_t)$?

In general, $q(x_{t-1}|x_t) \propto q(x_{t-1})q(x_t|x_{t-1})$ is intractable

Note that when β_t is small enough, $q(x_{t-1}|x_t)$ will be Gaussian

But we cannot easily estimate $q(x_{t-1}|x_t)$ it **needs entire dataset**.

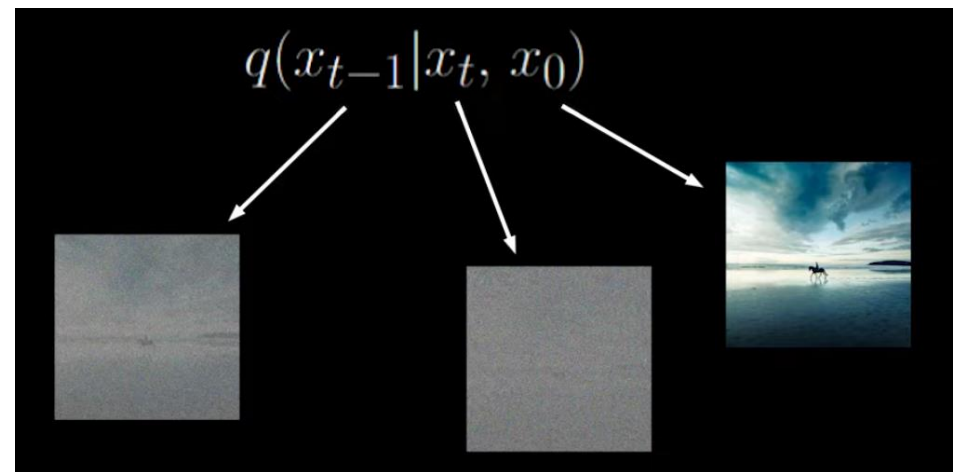
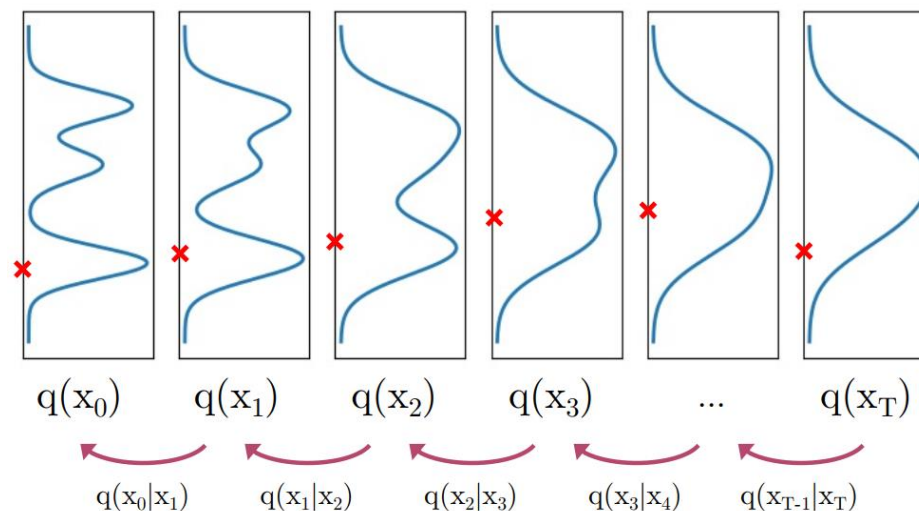
Generation:

Sample $x_T \sim N(x_T; 0, I)$

Iteratively sample $x_{t-1} \sim q(x_{t-1}|x_t)$

$q(x_{t-1}|x_t)$ is intractable; $q(x_{t-1}|x_t, x_0)$ is tractable

Diffused Data Distributions



Conditional forward and reverse process

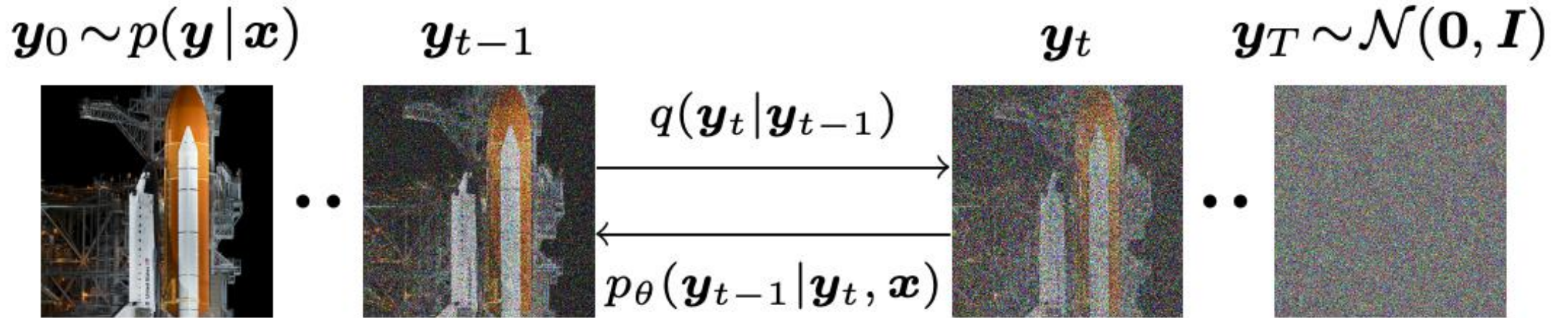


Figure 2: The forward diffusion process q (left to right) gradually adds Gaussian noise to the target image. The reverse inference process p (right to left) iteratively denoises the target image conditioned on a source image $\mathbf{x}$. Source image $\mathbf{x}$ is not shown here.

SR performance of SR3

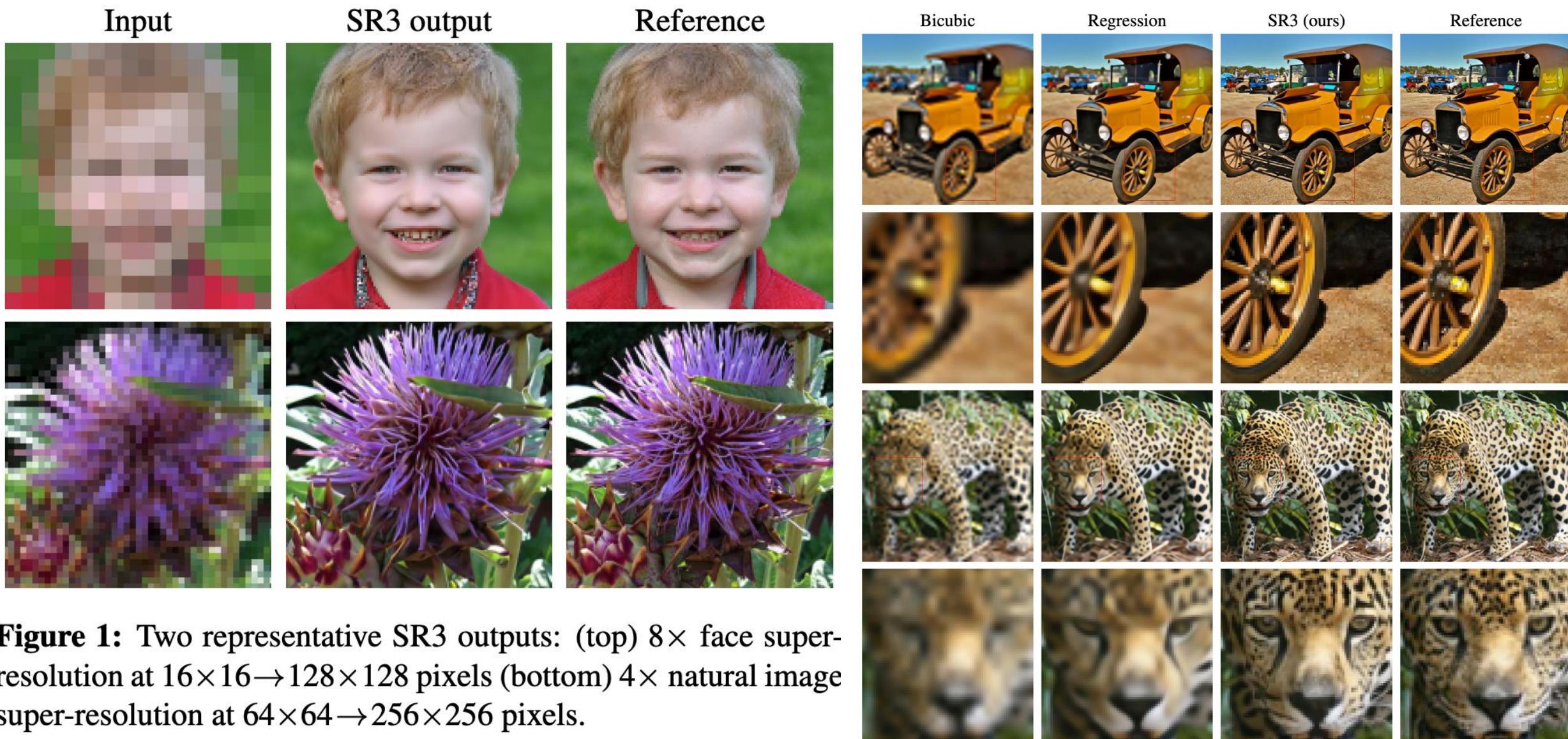
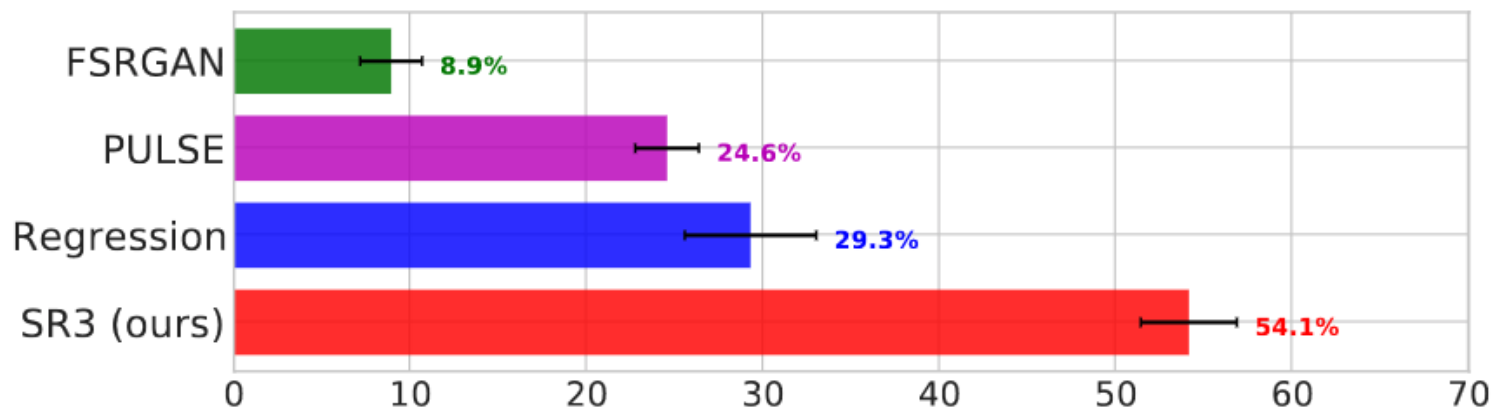


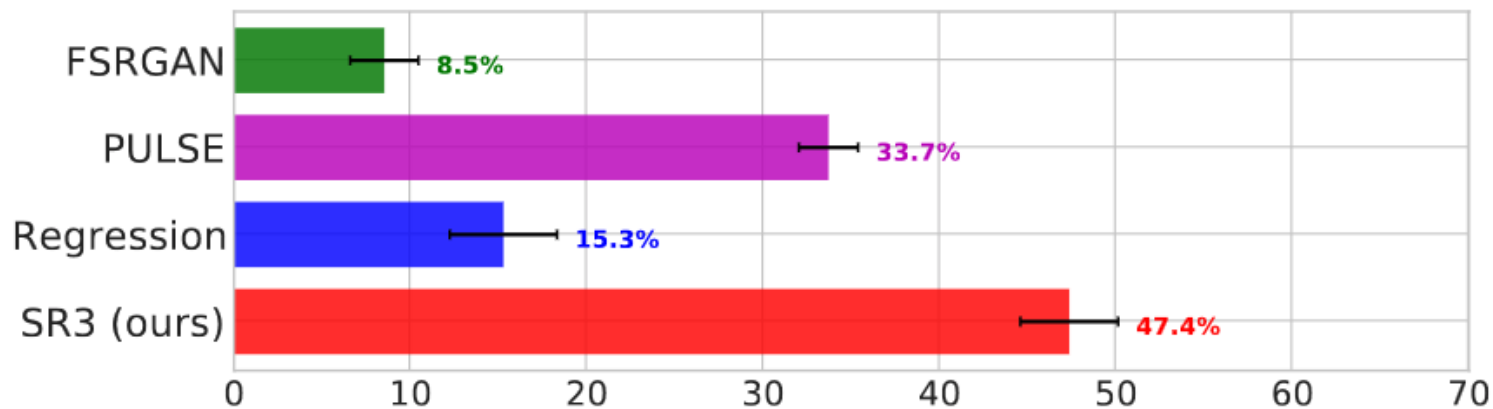
Figure 1: Two representative SR3 outputs: (top) $8\times$ face super-resolution at $16\times 16 \rightarrow 128\times 128$ pixels (bottom) $4\times$ natural image super-resolution at $64\times 64 \rightarrow 256\times 256$ pixels.

SR performance of SR3

Fool rates (3 sec display w/ inputs, $16 \times 16 \rightarrow 128 \times 128$)



Fool rates (3 sec display w/o inputs, $16 \times 16 \rightarrow 128 \times 128$)



IDM: Implicit Diffusion Models for Continuous Super-Resolution

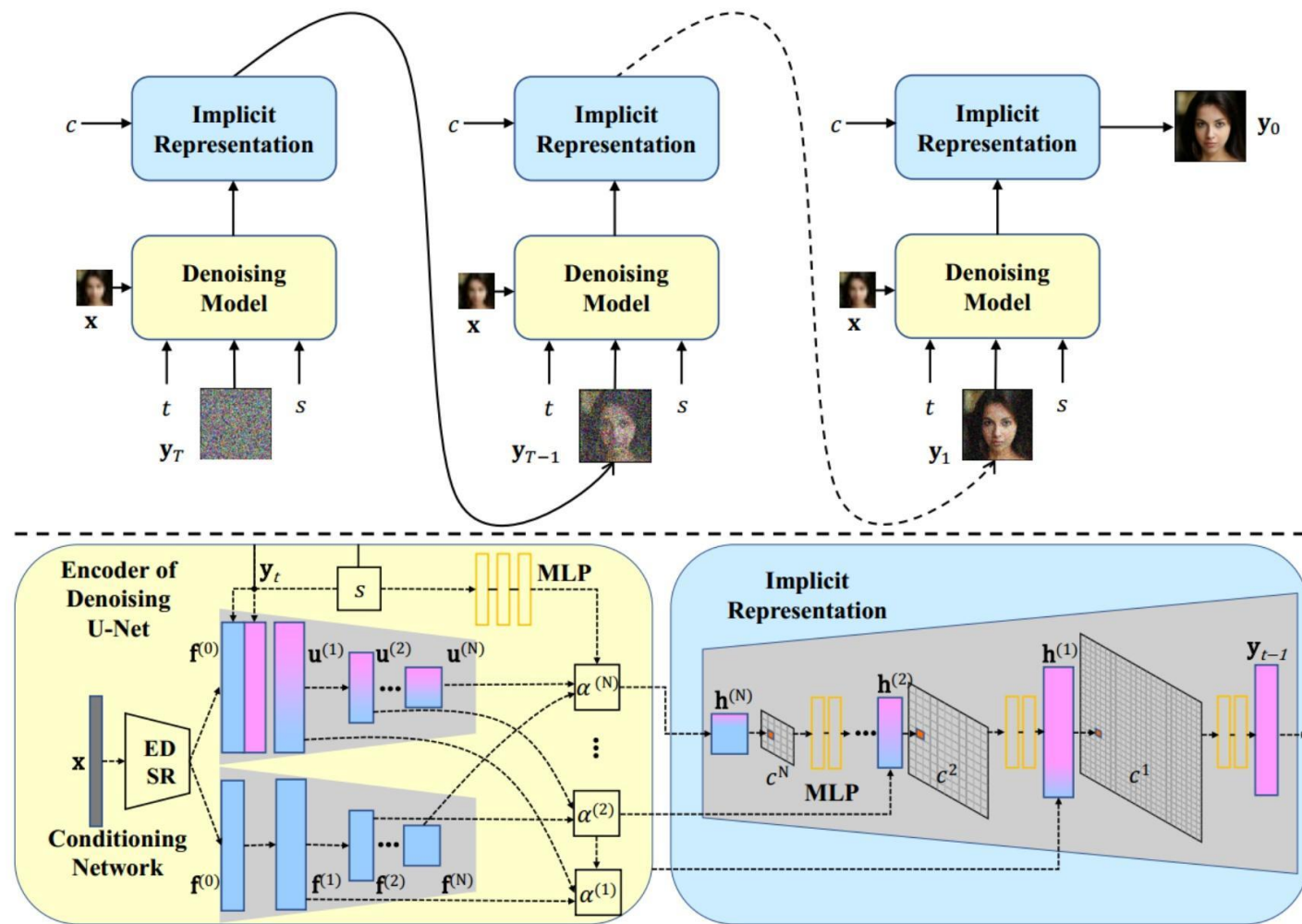


Figure 3. Overview of the IDM framework. **Upper Part:** Overall process of the inference. **Lower Part:** Detailed illustration of a denoising step, where the U-Net decoder is omitted for conciseness.

Performance of IDM

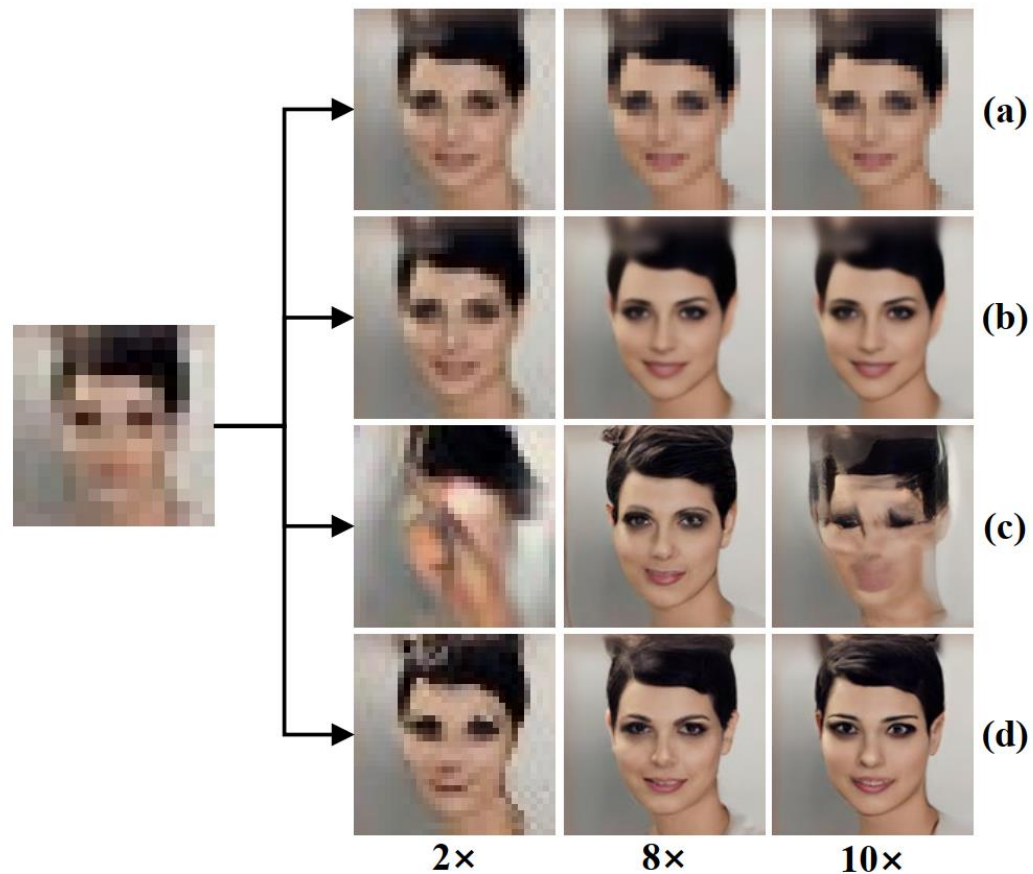
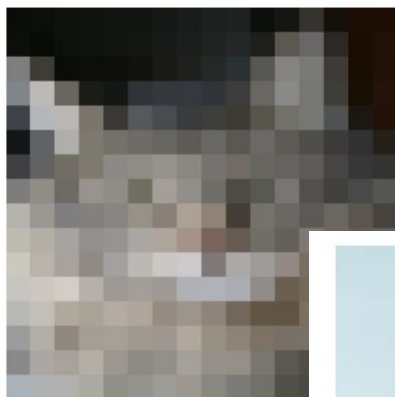


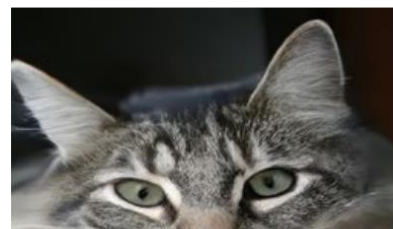
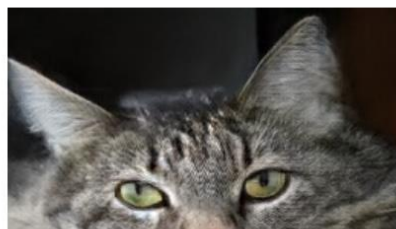
Figure 1. Visual comparison, where training is on $8\times$ SR and testing is on $2\times$, $8\times$, and $10\times$. (a) EDSR [23] and (b) LIIF [6] are regression-based models; (c) SR3 [35] and (d) IDM (ours) are generative models. Among them, LIIF and IDM employ the implicit neural representation.

Performance of IDM



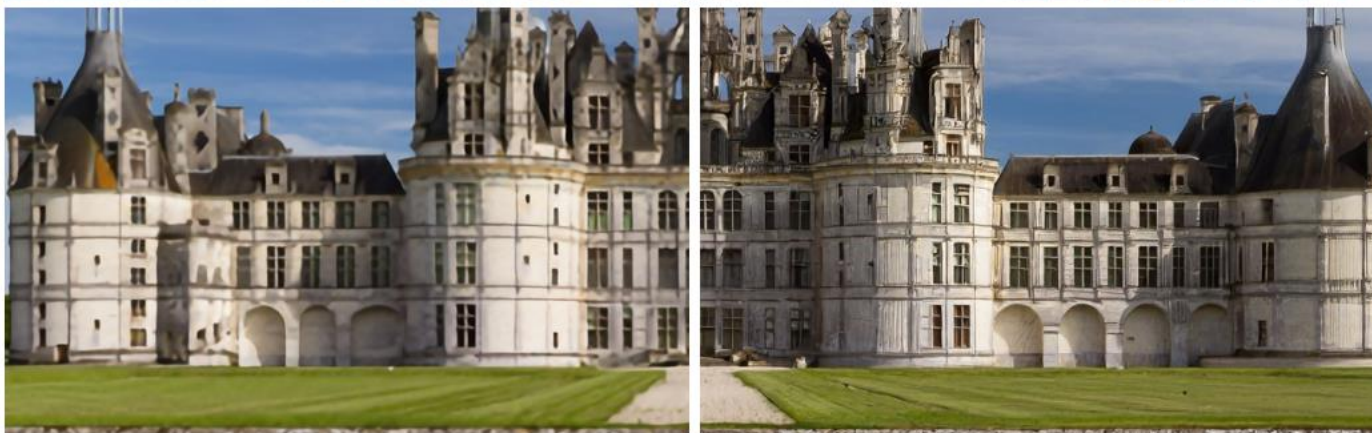
(a) Low-Resolution

Figure 2. Examples of inherent information, textures and GAN-specific details in the output with high-resolution.



(b) Ground-Truth

re GAN, and loses the
ill generates unnatural
anism, IDM generates



Take home message

- Learning high-frequency image content is difficult for supervised neural network paradigm. There are two ways to improve:
- Unsupervised methods to overfit the high-frequency image content in anisotropic image.
- Generative methods learn the latent space consistency between low-resolution and high-resolution image, and avoid from generating perceptual incorrect high-frequency texture.